Your Title*

Your Name

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Abstract

This is the abstract. An abstract is a short summary that tells a potential reader to quickly what you are doing allowing him to judge whether to read your paper. (abstract length is $\approx$ 100 words for a normal length paper)

1. Introduction

This document can be used as a template (in this case you want to delete all my text from here to the `\bibliographystyle{chicago}` command – you can keep some section headings if you like). In the following, I will give some examples of how to cite and reference, and how to use mathematics in LaTeX.[1] I assume that you have read the "Getting started" part of the LaTeX wikibook, see `https://en.wikibooks.org/wiki/LaTeX`.

1.1. Literature Review

Let me give some examples how to cite in LaTeX using BibTeX: The "Chicago view" of privacy is based on Stigler (1980) and Posner (1981). A page in a book is cited as Solove (2010, p. 34).

Note that all the cited articles, books etc. have to have an entry in the .bib file where you define the label used for citing and store all the information necessary for creating the reference list at the end of the document. You can usually copy and paste the entries for the .bib file from Google Scholar or JSTOR (and otherwise you do it by hand following Martin Osborne's guide; `https://www.economics.utoronto.ca/osborne/latex/BIBTEX.HTM`). To tell LaTeX that there is a .bib file, you need to use the `\bibliography{name .bib file` command at the point where you want to have the reference list; see the end of this document. The layout of the reference list is determined by the `\bibliographystyle{}` command; in this document I use the style "chicago".

*If necessary, you can – at this very point – thank someone for his great comments to earlier versions of the paper.

[1] Ideally, you should look at the .pdf and the .tex file at the same time to see what the code generates.

The advantage of BibTeX is twofold: First, you spell the author names and the publication years correctly. Second, the reference list is automatically generated and formatted nicely.

Do not forget to run BibTeX (unless you like question marks instead of citations all over your document). To get the full output:

1. run LaTeX
2. run BibTeX
3. run LaTeX twice.[2]

2. Maths

Mathematics within a line of text is put between $ marks. For example, `$\frac{3}{4}$` yields the fraction $\frac{3}{4}$ or $\lim_{x\rightarrow \infty}e^{-x}=0$ is generated by `$\lim_{x\rightarrow \infty}e^{-x}=0$`. Subindeces as in x_1 are written as `$x_1$`. Here is how you use math environments:

Definition 1 (Fixed Point)**.** *Let $f : D \to R$ where $D \subseteq R$. If $f(x) = x$, then x is called a* fixed point *of f.*

You reference the definition as "definition 1" (that is `definition \ref{def:fp}` in the source file where `def:fp` is the label of the definition above). Similarly, you can reference sections as `section \ref{sec:intro}` which yields "section 1". Also subsection 1.1 can be referenced using `subsection \ref{sec:literature}`. The advantage of such references over writing "section 1.1" directly is that your references will automatically adjust if you change something in the structure of your document; for example, if you insert another section. (Note that you have to compile LaTeX twice before it gets a new reference properly formatted.)

Important (or long) equations can be displayed in an own line (and maybe numbered and referenced) like so:

$$x^2 + px + q = 0,$$

$$x_{1/2} = \frac{p}{2} \pm \sqrt{(p/2)^2 - q}. \tag{1}$$

2.1. Results

A mathematical result, like a proposition or a lemma, is written in a similar environment as a definition.

[2] You do not have to run BibTeX again afterwards unless you insert a new citation.

Proposition 1. *Let $f : [0,1] \to [0,1]$ be an increasing function. Then, f has a fixed point.*

3. Extensions

This section gives examples of specialized uses that are probably unimportant if you try LaTeX for the first time.

3.1. Importing graphics

You can include a graphic with the command `\includegraphics{filename}` if the graphic is in the same folder as your .tex document (recommended file formats are png, eps or pdf). Usually it is a good idea to place the graphic in a "figure environment" as here.

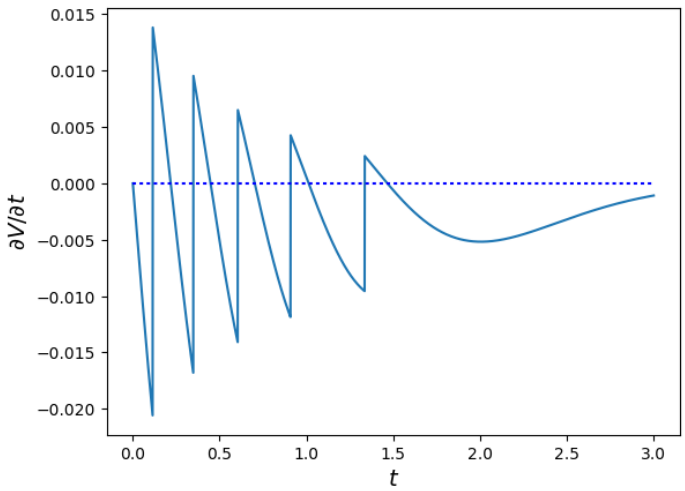

Figure 1: A nice figure

We can reference the figure as figure 1 using its label. Note that LaTeX will place the figure at the top of the next page if there is not enough space on the current page (while moving the text that seemingly follows the figure in the .tex file to fill up the remaining space).

3.2. Tables

Sometimes you need to have a nicely formatted table. I will give only a simple example and you can check `https://en.wikibooks.org/wiki/LaTeX/Tables` if you need something more complicated. The table itself is created in a "tabular environement". In order to give the table a caption and to be able to reference it, the tabular environment is put into a "table environment".

3.3. Generating graphics

If you need to create nice looking graphs, something like figure 1 for example, you can produce these inside your LaTeX document using a package called "tikz" (do not forget to

top left	top center	top right
mid left	mid center	mid right
bottom left	bottom center	bottom right

Table 1: A simple table

add `\usepackage{tikz}` to the preamble if you want to try). Jacques Crémer wrote an excellent introduction to tikz; see `http://cremeronline.com/LaTeX/minimaltikz.pdf`.

3.4. More maths

Putting two equations nicely formatted one below the other.

$$\alpha_h^+ = \frac{\alpha p}{(1-\alpha)/2 + \alpha p} \tag{2}$$

$$\alpha_h^- = \frac{\alpha(1-p)}{(1-\alpha)/2 + \alpha(1-p)}. \tag{3}$$

Sometimes an equation is too long to be displayed on a single line.

$$\frac{\alpha p}{(1-\alpha)/2 + \alpha p} - \frac{3}{4}\left(\frac{\beta+\gamma}{\rho} + \theta e^{\eta}\right)^3 + (\frac{\beta+\gamma}{\rho} + \theta e^{\eta})^3 = \sqrt{x} + \mathbb{E}_\theta\left[\log(\theta) * \sigma^2\right] + \frac{3+1}{x} + \frac{\alpha - \frac{2+p}{5}}{(1-\alpha)/2 + \alpha(1-p)}. \tag{4}$$

The previous equation also illustrates when to use normal paranthesis like "(...)" and when you need the command "`\left(...\right)`" (the latter adjusts the size of the paranthesis).

If you do not reference and equation later on, there is no point in having it numbered.

$$\begin{aligned} \alpha_h^+ &= \frac{\alpha p}{(1-\alpha)/2 + \alpha p} \\ \alpha_h^- &= \frac{\alpha(1-p)}{(1-\alpha)/2 + \alpha(1-p)} \end{aligned} \tag{5}$$

and for a single equation

$$\alpha_h^+ = \frac{\alpha p}{(1-\alpha)/2 + \alpha p}.$$

The following symbols are occasionally used: $\sim$, $\equiv$, $\approx$, $\Rightarrow$, $\Leftrightarrow$, $\succeq$, $\preceq$, $\succ$, $\prec$, $\nearrow$, $\Re$, $\mathbb{R}$, $\in$, $\emptyset$

3.5. Special characters and symbols

In case there are problems with typing special characters directly, here is how you can generate them directly:
Sørensen, æ, å, Øster, Å, Æ
Müller, ö, ä, Öfter, Ü, Ä
é, è , ô, ç, ą
45°, $, &

4. Conclusion

All scientific writing should be done in LaTeX.

Appendix

Proof of proposition 1: Let

$$A = \{x : \ x \geq f(x)\}. \tag{6}$$

Note that $0 \in A$ and therefore $A \neq \emptyset$. Let $x^* = \sup A$. Then $f(x^*) \geq x^*$ by the definition of A, see equation (6), and because f is increasing.

Also $f(x^*) \leq x^*$: if not, i.e. $f(x^*) > x^*$, then $x^* + \varepsilon > f(x^* + \varepsilon)$ for some $\varepsilon \in (x^*, f(x^*)] \cap [0, 1]$ because f is increasing which contradicts the definition of x^*. Hence, $x^* = f(x^*)$. $\square$

References

Posner, R. A. (1981). The economics of privacy. *American Economic Review 71*(2), 405–409.

Solove, D. J. (2010). *Understanding Privacy.* Cambridge, MA: Harvard University Press.

Stigler, G. J. (1980). An introduction to privacy in economics and politics. *Journal of Legal Studies 9*(4), 623–644.